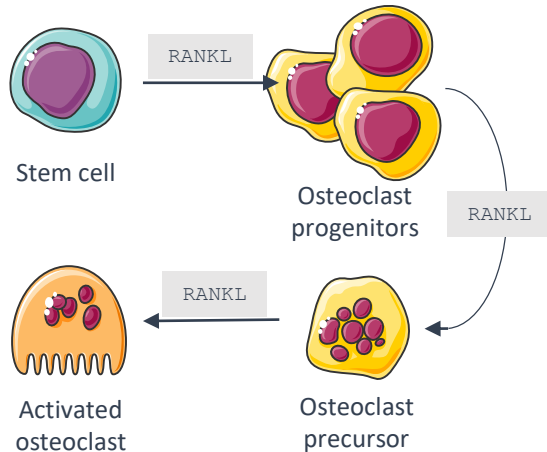


BONE REMODELING STAGES

Bone remodeling ensures the repair of micro-damage in bones, supports calcium and phosphate balance, and adapts bone structure to mechanical stress.

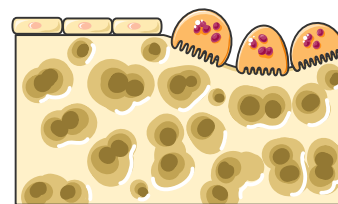
Step 1: Activation

Precursor cells are recruited to the remodeling site where bone resorption will occur.



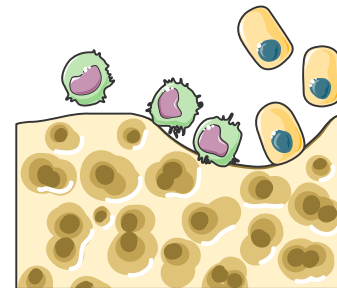
Step 2: Resorption

Osteoclasts break down the bone matrix, creating small pits on the bone surface.



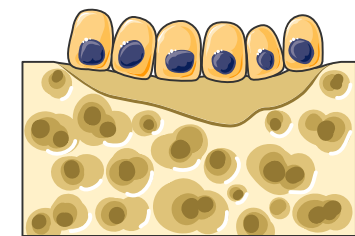
Step 3: Reversal

After resorption, a reversal phase occurs where precursor cells begin to lay down new bone.



Step 4: Formation

Osteoblasts build new bone matrix by depositing collagen and minerals, filling the resorbed pits.



RANKL (Receptor Activator of Nuclear Factor Kappa-B Ligand):
A signaling molecule that activates osteoclasts.

Educational Tools

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